

software platforms is the key strategy for long-term supportability and maintainability," cites Semancik.

"Adopting open standard instrumentation interfaces ensures not only that users have the ability to select from a wide range of COTS solutions but also that these solutions are forward-looking as well as backward-compatible," he adds.

"Increase in multifunctional and high-capacity products and preparing for the future products seem to be a tough focus. Solutions for different expectations are the main aspects to be addressed all time. Innovative techniques in handling T&M equipment by engineers should be adopted overall as well," Naresh adds.

Today's systems are convergence of many technologies. Let's take the example of a communication device that supports multiple radio standards like Bluetooth and Wi-Fi. "Designers are looking for complete solutions in an automated mode, but it is difficult to realise in the absence of standard test routines," explains Valsalan. Choosing the right combination of equipment that are properly synchronised is necessary.

"Equipment availability is sometimes an issue. High-performance test equipment must be readily available to test the systems across the lifetime. Ongoing, long-term support must also be affordable, and in line with the shifts between programme and operational budgets," shares Awasthi.

"High-end technology at an affordable cost is also a challenge on both the ends. Equipment also need to be light-weight so that more of it can be accommodated for the same lifting power. Efficiency of the equipment should help to reduce the operating cost for profitability," says Sabapathi. Apart from this, the T&M equipment should have less metallic content to avoid detection by enemy radars.

What's in store?

A lot of new developments are happening in the test and measurement space to support the technologies in-

vented in the aerospace and defence. "Technology is being used to provide virtually real-time intelligence to the aerospace and defence personnel, to interconnect them as never before and to develop reliability that reduces the risk to them," explains Awasthi.

The growing use of a platform-based approach in both instrumentation and software provides several advantages such as lower prices across families of products, better scalability of those products and simplified support.

One key benefit of software platforms is measurement consistency. For example, Awasthi says, "The new vector signal analysis (VSA) software can run inside a signal analyser, oscilloscope or logic analyser to analyse baseband, IF and RF, or digital signals, respectively. It can also be used to measure simulated signals within the Agilent Advanced Design System (ADS) software, bridging the gap between simulated and physical implementations of a new design." Because measurement results are consistent across platforms, these can be compared directly, facilitating troubleshooting and validation.

The PXI Express digitiser sets a new milestone for the performance of PXI modular instrumentation. With a bandwidth greater than 3 GHz, sample rates beyond 10 GS/s, data throughput of more than 600 MB/s and multi-module synchronisation capabilities, it is ideal for the aerospace and defence industry.

"Manufacturers take advantage of new multicore processors, FPGAs, faster fabrics like PCI Express, real-time processing, terabytes of data storage and networking to provide users with high-performance test and measurement solutions. With new testing developments, you can target diverse application areas such as software-defined radio, sensor data fusion and radar scan conversion. With an open, customisable signal front-end, the exact requirements of a

test or embedded system can be met," shares Krishnan.

Manufacturers are also looking at completely integrated signal conditioning and integrated calibration capabilities. Semancik explains, "This functionality eliminates the need to disconnect instrumentation from field terminations for calibration, and provides a convenient approach to calibrate the equipment immediately prior to test execution, resulting in the most accurate measurements possible."

Recently announced PXI Multi-Computing (PXImc) specification, by PXI Systems Alliance, supports a high-performance communication architecture for multi-controller PXI and automated test systems. It provides vendor-interoperable communication between multiple controllers or systems via a high-throughput, low-latency communication link.

Embedded Web interface is another emerging technology not available in previous generations of instrumentation. It provides the users with immediate out-of-the-box operation without any programming or other software applications. The embedded Web interface simplifies start-up operations and can provide significant time savings when verifying connections and set-up or debugging.

Technologies like field-programmable gate arrays, complex programmable logic gates and multi microprocessor environments have found their place in the new equipment. Software for virtual instrumentation, synthesis, data capture and analysis such as NI LabView, MATLAB, AWR, Agilent's ADS, VEE and VSA have also become a part of integrated testing systems. Bus interfaces like LXI are gaining popularity. All the equipment are designed and developed to give highly accurate and highly reliable results as this industry demands right data first time, every time. ●

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